How Much Should We Trust the Dictator's GDP Growth Estimates?

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Importance of the Quality of GDP Statistics

- GDP is one of the most widely used indicators of economic performance
- Even very poor countries report GDP statistics
- It is central to answering key questions in economics about the effectiveness and importance of various policies and factors:
    - Does democracy positively impact the economic growth? (Acemoglu et al., 2019)
    - Are institutions important for economic activity? (Acemoglu et al., 2001)
- GDP data can be noisy, but as long as they are not systematically biased by governments or institutions, they can still be quite useful.

Why GDP May Be Biased

- There are strong reasons to suspect that GDP statistics may be biased
- Governments have incentives to present the economy in a favorable way:
  - signal competence
  - influence public perception and political support
- In democracies:
  - media, opposition, and institutions constrain manipulation
- In autocracies:
  - weaker scrutiny and fewer checks
  - There exist both incentives and ability to manipulate data

Core Idea

- GDP is potentially manipulable, as it is reported by governments
- To address this, we need an alternative measure of economic activity that is not under government control
- Night-time lights (NTL) provide such a measure:
  - recorded by satellites, independent of governments statistical offices
  - reflects energy usage and is thus strongly correlated with economic activity
  - consistently measured across countries
- Comparing GDP to NTL allows us to detect potential manipulation

Empirical Model

$$\ln(GDP)_{i,t} = \mu_i + \delta_t + \phi_0 \ln(NTL)_{i,t} + \phi_1 FiW_{i,t} + \phi_2 FiW_{i,t}^2 + \phi_3(\ln(NTL)_{i,t} \times FiW_{i,t}) + \xi_{i,t}$$

- $\ln(GDP)_{i,t}$: reported GDP, country i, year t (World Bank).
- $\ln(NTL)_{i,t}$: night-time lights.
- $FiW_{i,t}$: Freedom in the World index (Freedom House), ranging from 0 to 6, where higher values indicate more autocratic regimes
- μ_i: country fixed effects
- δ_t: year fixed effects

The Interaction Term

- The term $\ln(NTL)i,t \times FiWi,t$ allows the elasticity between GDP and NTL to vary with the political regime.
- The marginal effect of NTL on GDP is:

$$\frac{\partial \ln(GDP)_{i,t}}{\partial \ln(NTL)_{i,t}} = \phi_0 + \phi_3 \cdot FiW_{i,t}$$

- If $\phi_3 > 0$, then:
  - the same increase in NTL is associated with a larger increase in GDP in more autocratic regimes
- This suggests that GDP is overstated in autocracies relative to true economic activity

Parameter of Interest

- The ratio
$$\frac{\phi_3}{\phi_0}$$
measures how the elasticity of GDP with respect to NTL changes with a one-unit increase in autocracy (FiW)
- To obtain the overall exaggeration, this effect is scaled by the interquartile range (IQR) of FiW, which gives us our parameter of interest:
$$\sigma \approx \frac{\phi_3}{\phi_0} \times IQR(FiW)$$
- This corresponds to comparing a typical democracy to a typical autocracy in the sample

Baseline Results

	Dependent variable: $\ln(GDP)_{i,t}$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\ln(NTL)_{i,t}$	0.296***	0.292***	0.215***	0.214***	0.251***	0.265***	0.288***
	[0.044]	[0.042]	[0.043]	[0.043]	[0.044]	[0.042]	[0.067]
$FiW_{i,t}$		-0.023**	-0.006	-0.015			-0.086*
		[0.010]	[0.010]	[0.025]			[0.048]
$FiW_{i,t}^2$				0.002			0.012
				[0.005]			[0.008]
$\ln(NTL)_{i,t} \times FiW_{i,t}$			0.021***	0.022***			0.032***
			[0.005]	[0.005]			[0.008]
$D(\text{Partially Free})_{i,t}$					0.001		
					[0.020]		
$D(\text{Not Free})_{i,t}$					0.014		
					[0.039]		
$\ln(NTL)_{i,t} \times D(\text{Partially Free})_{i,t}$					0.041***		
					[0.015]		
$\ln(NTL)_{i,t} \times D(\text{Not Free})_{i,t}$					0.067***		
					[0.021]		
$D(\text{Autocracy})_{i,t}$						0.043**	
						[0.018]	
$\ln(NTL)_{i,t} \times D(\text{Autocracy})_{i,t}$						0.047***	
						[0.011]	
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,895	3,895	3,895	3,895	3,895	3,895	332
Countries	184	184	184	184	184	184	166
(Within country) R^2	0.219	0.226	0.259	0.260	0.238	0.240	0.466
$\hat{\sigma}$			0.342	0.354	0.269	0.177	0.388
$\hat{\sigma}$ S.E.			(0.158)	(0.166)	(0.124)	(0.060)	(0.264)
$\hat{\sigma}$ 95% C.I.			[0.14,0.78]	[0.14,0.82]	[0.09,0.62]	[0.09,0.32]	[0.11,1.05]

Where Does Manipulation Occur?

Table 2: The Autocracy Gradient in the NTL Elasticity of GDP Sub-Components

	Consumption	Investment	Government	Exports	Imports
	(1)	(2)	(3)	(4)	(5)
$\ln(\text{NTL})_{i,t}$	0.184***	0.353***	0.210***	0.354***	0.253***
	[0.041]	[0.083]	[0.060]	[0.077]	[0.054]
$\text{FiW}_{i,t}$	-0.003	0.023	-0.002	-0.007	-0.006
	[0.035]	[0.062]	[0.041]	[0.058]	[0.042]
$\text{FiW}_{i,t}{}^2$	-0.002	-0.010	-0.001	-0.004	-0.005
	[0.006]	[0.012]	[0.007]	[0.011]	[0.008]
$\ln(\text{NTL})_{i,t} \times \text{FiW}_{i,t}$	0.004	0.040***	0.030***	0.011	0.013*
	[0.006]	[0.010]	[0.007]	[0.012]	[0.008]
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	3,416	3,414	3,416	3,416	3,416
Countries	173	173	173	173	173
(Within country) R^2	0.081	0.141	0.128	0.095	0.099
$\hat{\sigma}$	0.078	0.400	0.505	0.105	0.174
$\hat{\sigma}$ S.E.	(0.126)	(0.245)	(0.362)	(0.188)	(0.156)
$\hat{\sigma}$ 95% C.I.	[-0.15,0.38]	[0.15,0.96]	[0.20,1.45]	[-0.17,0.45]	[-0.02 0.53]

When Does Manipulation Occur?

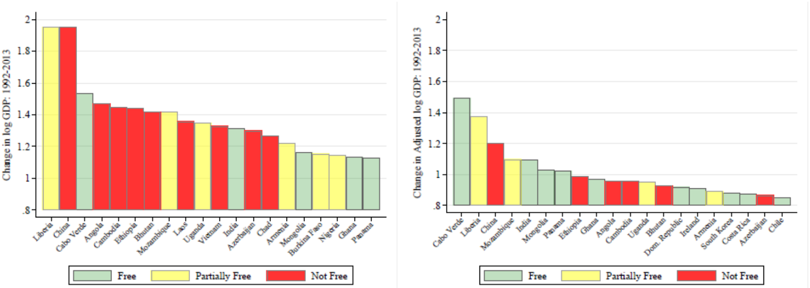
Table 3: Heterogeneous Effects: Economic Underperformance

	Dependent variable: $\ln(\text{GDP})_{i,t}$				
	(1)	(2)	(3)	(4)	(5)
$\ln(\text{NTL})_{i,t}$	0.280***	0.203***	0.214***	0.239***	0.241***
	[0.063]	[0.055]	[0.053]	[0.060]	[0.057]
$\ln(\text{NTL})_{i,t} \times \text{D(Low Growth)}_{i,t}$	0.003	-0.001	-0.018***	0.000	-0.011*
	[0.005]	[0.005]	[0.007]	[0.005]	[0.006]
$\ln(\text{NTL})_{i,t} \times \text{FiW}_{i,t}$		0.022***	0.016***		
		[0.005]	[0.005]		
$\ln(\text{NTL})_{i,t} \times \text{FiW}_{i,t} \times \text{D(Low Growth)}_{i,t}$			0.007**		
			[0.003]		
$\ln(\text{NTL})_{i,t} \times \text{D(Partially Free)}_{i,t}$				0.041***	0.035**
				[0.015]	[0.014]
$\ln(\text{NTL})_{i,t} \times \text{D(Not Free)}_{i,t}$				0.067***	0.046**
				[0.021]	[0.020]
$\ln(\text{NTL})_{i,t} \times \text{D(Partially Free)}_{i,t} \times \text{D(Low Growth)}_{i,t}$					0.012
					[0.009]
$\ln(\text{NTL})_{i,t} \times \text{D(Not Free)}_{i,t} \times \text{D(Low Growth)}_{i,t}$					0.034**
					[0.016]
Country FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	3,895	3,895	3,895	3,895	3,895
Countries	184	184	184	184	184
(Within country) R^2	0.219	0.260	0.267	0.238	0.247
$\hat{\sigma}$		0.374		0.281	
$\hat{\sigma}$ S.E.		(0.211)		(0.153)	
$\hat{\sigma}$ 95% C.I.		[0.13,0.95]		[0.09,0.73]	

Democracy and Economic Growth

- Autocracies do not grow faster
    - Raw GDP - Not free: 85%, Partially free: 76%, Free: 61%
    - Adjusted GDP - Not free: 55%, Partially free: 57%, Free: 56%

Figure 6: Top 20 Fastest-Growing Economies: 1992/3 - 2012/13



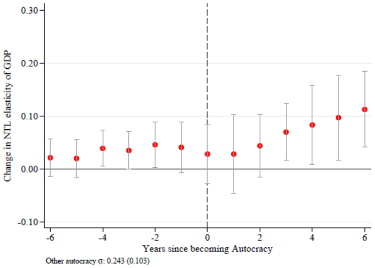
(a) Raw GDP

(b) Adjusted GDP

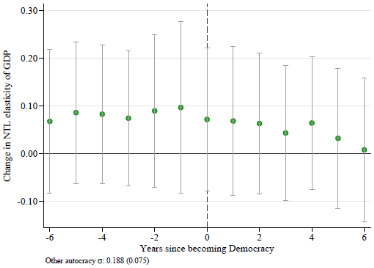
Democracy Does Cause Statistics to Be Better

- Studies of the effects of democratization (autocratization) on economic growth are biased downward (upward).

Figure 4: Political Transitions



(a) Democracy to Autocracy



(b) Autocracy to Democracy

Questions

Thank you